

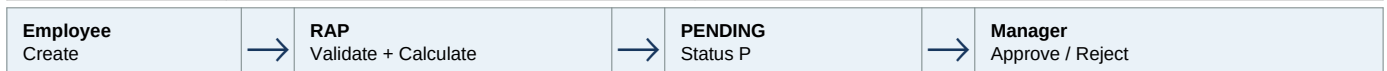
SAP RAP – Employee Leave Management

Technical Documentation | DEV Training Application

Purpose: A Fiori Elements leave-management application built with SAP RAP. Employees create leave requests, while managers approve or reject them. Approved leave automatically updates the employee leave balance.

1. Application Architecture

Layer	Project Object	Purpose
Data Model	ZLM_LEAVE_REQ ZLM_LEAVE_BAL ZLM_EMPLOYEE	Stores leave requests, balances and employee data.
RAP BO	ZR_LM_LEAVE_REQ ZR_LM_LEAVE_BAL	Managed RAP behavior with draft processing.
Service	ZUI_LM_LEAVE ZUI_LM_LEAVE_O4	Exposes the application through OData V4.
UI	ZC_LM_LEAVE_REQ Metadata Extension	Provides Fiori Elements list and object pages.



2. RAP Behavior and Business Rules

- **Create:** Leave ID is entered and a new request is initialized as Pending.
- **Calculate Days:** No. of Days = To Date – From Date + 1.
- **Date Validation:** Both dates are mandatory and To Date cannot be before From Date.
- **Employee Validation:** Employee ID must exist in the employee master.
- **Balance Validation:** Requested days must not exceed available balance.
- **Overlap Validation:** An employee cannot have overlapping approved leave.
- **Approve:** Only Pending requests can be approved; status changes P → A.
- **Balance Update:** UsedLeaves increases by NoOfDays and Balance decreases by NoOfDays.
- **Reject:** Only Pending requests can be rejected; status changes P → R and balance is unchanged.
- **Status Control:** Status is read-only for users and initialized by determination.

3. Main Technical Objects

Object	Role
ZR_LM_LEAVE_REQ	Root RAP behavior entity for leave request processing.
ZR_LM_LEAVE_BAL	RAP behavior entity for employee leave balances.
ZBP_R_LM_LEAVE_REQ	Behavior pool implementing actions, validations and determinations.
ZC_LM_LEAVE_REQ	Projection view used by the UI service.
ZUI_LM_LEAVE	Service definition exposing the projection.
ZUI_LM_LEAVE_O4	OData V4 service binding for the Fiori Elements application.

4. End-to-End Flow

CREATE Employee enters data	VALIDATE Rules are checked	PENDING Status = P	APPROVE P → A	BALANCE Used ↑ / Balance ↓
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5. Testing Summary

- TC01 Valid Leave – passed.
- TC02 Invalid Dates – validation implemented.
- TC03 Insufficient Balance – validation implemented.
- TC04 Overlapping Leave – validation implemented.
- TC05 Approve – passed; status and balance updated.
- TC06 Reject – passed; status changed to R and balance unchanged.
- TC07 Search / Filter – supported by Fiori Elements.
- TC08 Unauthorized Approval – authorization stage skipped for this DEV training project.

6. Deployment Status

DEV: Completed and demonstrated.

QAS: Not deployed in the training environment.

Production: Not deployed.

Note: No production deployment is claimed because the project was completed in a DEV-only training system.

Project Outcome: Working SAP RAP + OData V4 + Fiori Elements leave-management application with validations, automatic day calculation, approval/rejection workflow and leave-balance management.